

Hiver SDE Intern Take-Home Assignment

AmazonHelp AI Customer Support Agent — Technical Report

1. Overview

This report documents a prototype AI customer-support agent built for the AmazonHelp brand using the Customer Support on Twitter dataset. The agent (1) classifies incoming customer messages into a data-derived intent taxonomy, (2) retrieves historically similar support interactions to ground a draft reply, and (3) applies an escalation policy to decide between AUTO-HANDLE and ESCALATE, with a stated reason. The report emphasizes evaluation discipline: two leakage issues were identified during development — in intent evaluation and in retrieval evaluation — and corrected before final results were reported. Initial optimistic numbers are shown alongside the corrected, leakage-resistant results throughout.

2. Data Preparation

- Brand selected: AmazonHelp, chosen based on usable interaction volume in the dataset.
- Constructed customer → AmazonHelp support conversation pairs from the raw Twitter dataset.
- Historical conversation-pair dataset: 168,814 usable AmazonHelp interactions.
- Each pair contains: customer tweet ID, customer text, customer time, support tweet ID, support text, support time.

3. Intent Taxonomy

An 11-class, data-derived intent taxonomy was defined to cover the observed range of AmazonHelp customer requests, including a catch-all class for unclear or unsupported requests.

Intent	Description
I01 order_delivery	Order status, delayed delivery, estimated delivery date, delivery timing
I02 missing_package	Package marked delivered but missing, not received, delivery disappearance
I03 return	Returning an item, return eligibility, return process/instructions
I04 refund	Refund requested, refund status, refund amount or timing
I05 wrong_damaged_item	Wrong, damaged, defective, broken or unusable item
I06 payment_billing	Payment failure, unexpected charge, billing issue, payment method/card
I07 account	Account access, login, password, account settings
I08 prime_subscription	Prime membership, Prime benefits, subscription, Prime charges
I09 cancellation_modification	Order cancellation, changing an order/address/order modifications
I10 general_information	General product/service/feature/policy information
I11 other_unclear	Unclear, miscellaneous, or unsupported customer request

4. Golden Evaluation Set

A golden set of 200 manually labeled examples was created (required range: 150–250), labeled with intent, escalation decision, and escalation reason.

Intent	Count
I11 other_unclear	52
I01 order_delivery	52

Intent	Count
I10 general_information	26
I05 wrong_damaged_item	13
I07 account	12
I02 missing_package	10
I08 prime_subscription	10
I06 payment_billing	9
I03 return	6
I09 cancellation_modification	5
I04 refund	5

5. Intent Classification: Baselines, Leakage, and Final Model

5.1 Baselines

Two baselines were established prior to model comparison:

- Majority-class baseline (predicts I11_other_unclear): 26.0% accuracy, 10.73% weighted F1.
- TF-IDF + Logistic Regression (initial, non-OOF): 32.0% accuracy, 29.42% weighted F1.

5.2 Leakage Correction

LEAKAGE CORRECTION — INTENT EVALUATION

The initial 83% intent accuracy was optimistic because the classifier had access to the full golden set during training. I therefore replaced it with 5-fold out-of-fold evaluation, where each prediction is generated by a model that did not train on that example.

5.3 Final Model Comparison (5-fold stratified out-of-fold)

Model	Accuracy	Wtd. Precision	Wtd. Recall	Wtd. F1
Word TF-IDF + Logistic Regression	43.5%	37.81%	43.5%	39.54%
Word + Char TF-IDF + Logistic Regression	44.5%	41.63%	44.5%	39.04%
Word TF-IDF + Linear SVM	42.0%	36.33%	42.0%	36.92%

Selected model: Word TF-IDF + Logistic Regression, on weighted F1 — the primary metric given class imbalance. The Word + Character variant had marginally higher accuracy but lower weighted F1, so it was not selected.

6. Historical Retrieval

- TF-IDF retrieval index built over 168,814 historical AmazonHelp customer messages.
- Retrieval uses word n-grams with cosine similarity to find historically similar interactions and provide grounding evidence for drafting.

LEAKAGE CORRECTION — RETRIEVAL EVALUATION

The golden examples originated from the historical corpus, creating potential self-match leakage. Leakage analysis: 200 unique golden IDs, all 200 present in the historical corpus, 238 historical rows belonged to golden IDs, 32 golden IDs had duplicate historical rows. The final retrieval evaluation therefore excludes each golden customer_tweet_id from its own retrieval candidates.

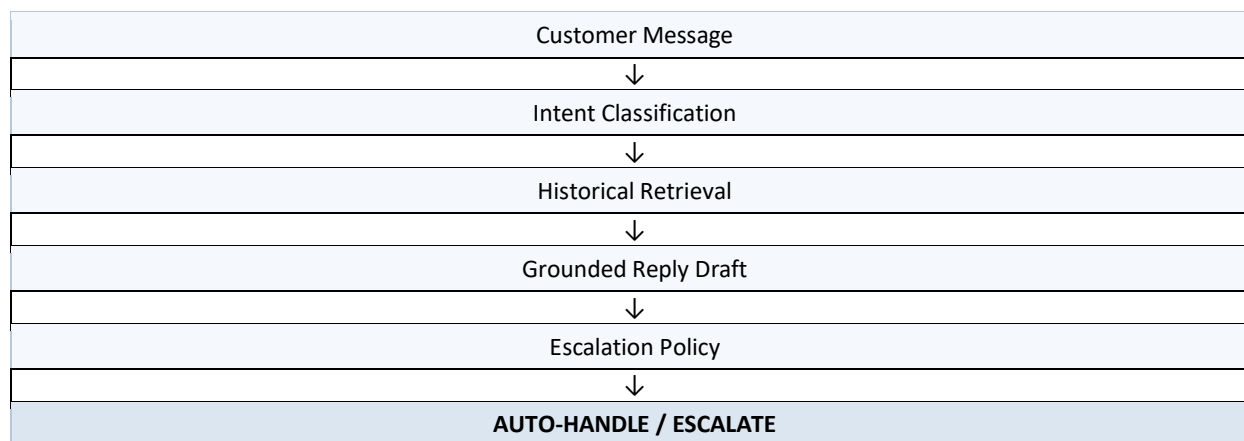
Final leakage-safe retrieval result (cosine threshold ≥ 0.20):

Metric	Value
Leakage-safe evidence availability	98.5%

Metric	Value
Mean similarity	0.4386
Median similarity	0.3715
Minimum similarity	0.0000

Note: this figure measures evidence availability, not end-to-end agent accuracy — see Section 10.

7. Support Agent Architecture



The draft response is grounded in a retrieved historical AmazonHelp support response rather than freely generated, keeping the draft traceable to precedent.

8. Escalation Policy and Results

8.1 Escalation Policy (V2)

The system escalates to a human when it detects:

- Explicit human request
- Sensitive account or payment issue
- Serious product issue
- Delivery failure or missing package
- Repeated unresolved issue
- Serious customer complaint
- No safe historical resolution based on low retrieval similarity

Otherwise, the system auto-handles when a relevant historical resolution is found.

8.2 Final Escalation Results (200-example golden set)

Metric	Value
Accuracy	69.5%
Precision	83.0%
Recall	42.4%
F1	56.12%
Auto-handle rate	76.5%
Missed escalations	53
Unnecessary escalations	8

Interpretation: the relatively high precision indicates that escalations are usually meaningful, but recall remains a weakness — the system still misses many interactions that should be escalated.

9. Failure Analysis

- **1. Missed escalations in order/delivery cases** — 19 missed escalations fell in I01_order_delivery. Hypothesis: relevant delivery history can cause the system to auto-handle even when human intervention is warranted.
- **2. Ambiguous / unclear requests** — 8 missed escalations fell in I11_other_unclear. Hypothesis: unclear messages make both intent and escalation decisions difficult.
- **3. Missing-package under-escalation** — 7 missed escalations fell in I02_missing_package. Hypothesis: retrieval finds similar delivery responses even when the individual situation requires escalation.
- **4. Retrieval similarity does not guarantee safe automation** — the 53 missed escalations had mean similarity 0.355 (median 0.340, max 0.746). Hypothesis: semantic similarity measures historical resemblance, not whether the retrieved resolution is safe or appropriate for the current case.
- **5. Unnecessary escalations** — 8 cases, spanning general information, payment/billing, order/delivery, unclear, and product issues. Hypothesis: safety-oriented escalation rules can be overly conservative in borderline cases.

10. What is misleading about my headline number?

- **Intent accuracy:** the initial 83% intent accuracy was misleading because it used evaluation examples during training. The defensible headline is the 43.5% out-of-fold accuracy / 39.54% weighted F1 (Section 5).
- **Retrieval:** the 98.5% figure measures evidence availability / retrieval success, not end-to-end agent correctness, resolution correctness, or escalation correctness (Section 6).
- **Reply quality:** the 4.31/5 human reply-quality score comes from only 20 manually rated examples, so it is a validation signal rather than a definitive production-quality estimate (Section 11).

11. Human Reply Quality Evaluation

20 examples were manually rated on a 1–5 scale across four dimensions.

Dimension	Mean Score (/5)
Groundedness	5.00
Resolution correctness	4.00
Relevance	4.55
Helpfulness	3.70
Overall	4.31

Rating distributions — Groundedness: 5×20. Resolution correctness: 2×1, 3×4, 4×9, 5×6. Relevance: 2×1, 3×2, 4×2, 5×15. Helpfulness: 1×1, 2×2, 3×2, 4×12, 5×3.

11.1 LLM-as-Judge Limitation

LLM-AS-JUDGE — NOT COMPLETED

LLM-as-judge evaluation was attempted using a local Qwen2.5 model, but the available CPU-only runtime did not allow reliable completion of the required 20-example evaluation. Therefore, LLM-generated scores and human–LLM agreement statistics are not reported. Reply quality is instead reported using the completed 20-example human evaluation.

12. Engineering Decision Log

#	Decision
1	Selected AmazonHelp based on usable interaction volume
2	Built customer → support conversation pairs
3	Used 11 intents
4	Added Other/Unclear catch-all intent
5	Created a 200-example golden set
6	Used stratified out-of-fold intent evaluation
7	Selected TF-IDF + Logistic Regression
8	Added historical retrieval
9	Prevented retrieval self-match leakage
10	Added explicit human-request escalation
11	Added sensitive account/payment escalation
12	Added serious/repeated issue escalation
13	Added retrieval safety threshold
14	Kept escalation policy V2 frozen to avoid overfitting
15	Used human reply-quality evaluation

13. Limitations

- Intent classification is weak for rare intents due to only 200 labeled examples and class imbalance.
- OOF intent accuracy is 43.5% and weighted F1 is 39.54%.
- Escalation recall is only 42.4%.
- Retrieval similarity alone is not sufficient to establish safe automation.
- Human reply evaluation contains only 20 examples.
- LLM-as-judge could not be completed reliably in the available CPU-only environment.
- The system does not execute refunds, modify orders, access customer accounts, or make autonomous business decisions — it only classifies, retrieves evidence, drafts responses, and recommends auto-handle vs. escalation.

14. One-Week Improvement Plan

- Expand the golden set, especially for rare intents.
- Improve intent classification with stronger semantic / few-shot approaches.
- Separate retrieval confidence from automation safety.
- Optimize escalation for higher recall on sensitive and unresolved cases.
- Improve reply generation using top-3 retrieved cases and extracted common resolution patterns rather than copying a single historical response.
- Run a reliable hosted LLM-as-judge evaluation and calculate human-vs-LLM agreement.

15. Final Results Summary

Component	Metric	Result
Intent classification	OOF Accuracy	43.5%
Intent classification	Weighted F1	39.54%

Component	Metric	Result
Retrieval	Leakage-safe evidence availability	98.5%
Escalation	Accuracy	69.5%
Escalation	Precision	83.0%
Escalation	Recall	42.4%
Escalation	F1	56.12%
Automation	Auto-handle rate	76.5%
Human reply quality	Overall score	4.31 / 5

15.1 Baseline Comparison

System	Accuracy	Weighted F1
Majority baseline	26.0%	10.73%
TF-IDF + Logistic Regression baseline	32.0%	29.42%
Word + Character TF-IDF (candidate)	44.5%	39.04%
Word TF-IDF + Logistic Regression (selected)	43.5%	39.54%
Word TF-IDF + Linear SVM (candidate)	42.0%	36.92%

16. Conclusion

This prototype demonstrates the complete support-agent pipeline for AmazonHelp end-to-end — intent classification, leakage-safe historical retrieval, grounded reply drafting, and an escalation policy — evaluated with explicit correction of two leakage issues rather than reporting optimistic, uncorrected numbers. Escalation recall and intent classification accuracy remain the primary areas for improvement before the system could be considered for further development toward production use.